<Empowering Rural Agriculture with Edge Computing>

D4E Assignment

Use case: < Empowering Rural Agriculture with Edge Computing>

Scenario: Empowering Rural Agriculture with Edge Computing

The small farmers throughout rural India including Prakash experience multiple obstacles stemming from poor resource governance, weak communications and inadequate data availability for moment-to-moment critical choices. Local edge computing and IoT sensors enable the described system to process agricultural data onsite for real-time investigation that improves irrigation management and pest control and fertilizer application decision-making. Data processing at local levels via this system reduces both cloud infrastructure dependency and enhances resource performance and shortens data delay times. The solution provides farmers from all locations with equal access to contemporary technology through its distribution so they can participate in fair and ecologically beneficial agricultural work. Edge computing integration gives farmers in underserved regions better agricultural outcomes through higher yields and reduced environmental impact and improved operational efficiency that leads to better farm industry conditions.

Overview of edge computing in the agricultural Internet of Things: key technologies, applications, challenges. (2020). IEEE Journals & Magazine | IEEE Xplore. https://ieeexplore.ieee.org/document/9153160

Use case: < Empowering Rural Agriculture with Edge Computing>

Main tasks of the system:

Data Collection: The farmer uses IoT sensors to obtain measurements of soil moisture along with temperature conditions and crop health status from different areas of the property.

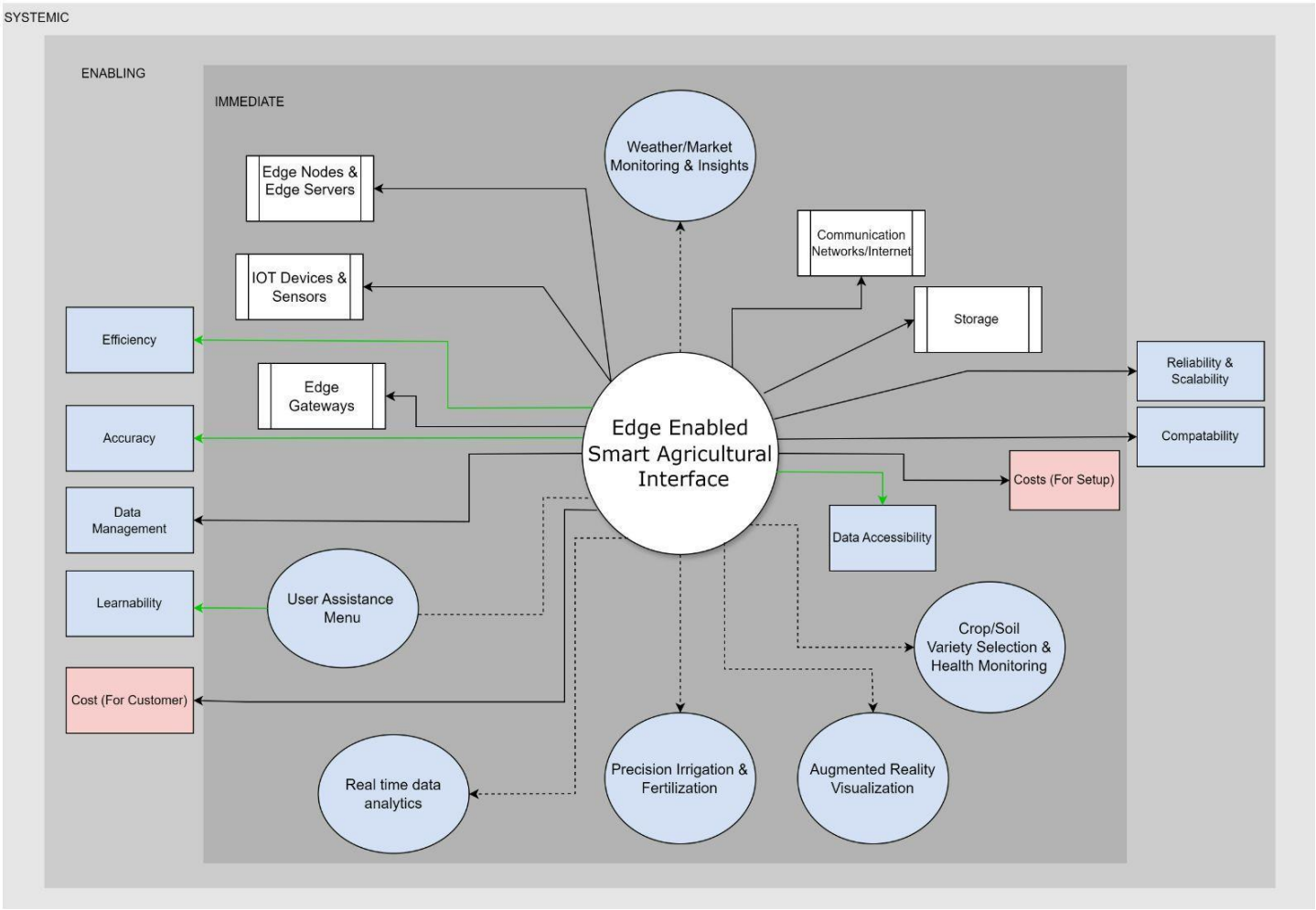
Edge Computing: Edge devices handle the data processing operation locally on-field to provide instantaneous decisions with short response times.

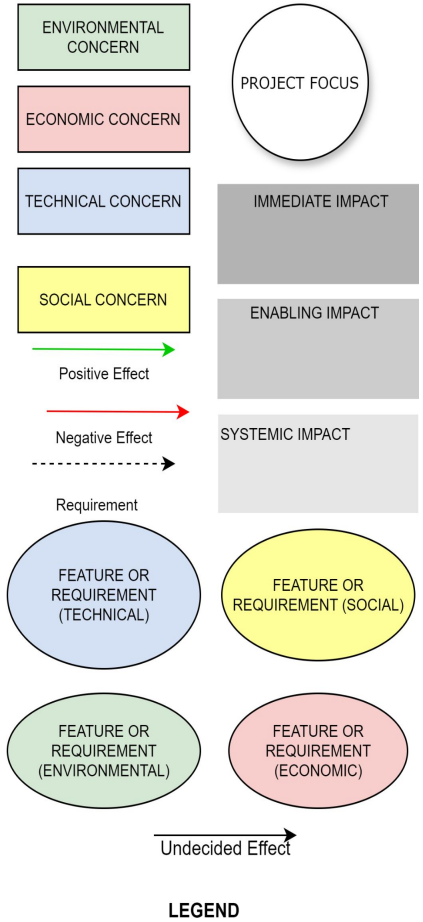
Real time Predictions and Recommendations: Prakash receives direct recommendations about irrigation actions and pest management together with fertilization advice through notifications generated by the system from the collected data.

Visualization: The augmented reality (AR) technological feature enables Prakash to view data overlays that present information about soil moisture and crop vitality while he is in his field.

Optimization: Through system recommendations Prakash can achieve optimal use of water and fertilizer as well as pesticides thus obtaining better crop yield alongside sustainability improvements.

Definition of Fairness: Decision Map





Avanija, J., Rajyalakshmi, C., Madhavi, K. R., & Rao, B. N. K. (2024). Enabling Smart Farming Through Edge Artificial Intelligence (AI). In Agriculture and Aquaculture Applications of Biosensors and Bioelectronics (pp. 69-82). IGI Global Scientific Publishing.

Lago, P., Condori Fernandez, N., Fatima, I., Funke, M., & Malavolta, I. (2024). The sustainability assessment framework toolkit: a decade of modeling experience. *Software and Systems Modeling* , 1-23.

Definition of Fairness: Decision Map (cont.)

The decision map provides an extensive Edge-Enabled Smart Agricultural Interface which features multiple essential components to enhance agricultural operations. Real-time weather condition and soil characteristic and crop health details are obtained by sensors and edge nodes through synced data collection from weather/market monitoring systems. Through precision irrigation and precision fertilization tools farmers achieve best resource management by reducing wastage and conserving natural ecosystems. Through its data accessibility system farmers of every region can access vital information without Internet limitations and thus experience decreased digital gaps. Augmented reality visualization tools enable farmers to make sense of complex data yet user assistance systems support complete system usage for high usability and easy learnability. Cost management occurs because the system chooses inexpensive infrastructure solutions that combine edge and cloud computing functions. The system features scalable attributes and dependable characteristics that allow its expansion with the growing agricultural sector together with reliable operation. The merger of ethical and collaboration-based design approaches results in fair sustainable decision-making procedures that let smallholders benefit from edge computing and IoT technologies.

Definition of Fairness: SQ model

<https://docs.google.com/spreadsheets/d/1FueEFMz3RcKaB2feQTvwa1M5oiWR7xgUDmSjVXYRjOU/edit?gid=0#gid=0>

Sustainability Dimension of the QA						
Quality Attribute	Definition	Reference	Technical	Environmen	Economic	Social
Data Accessibility	Ensures equal access to advanced technologies like IoT sensors and edge computing for all farmers.	[ISO/IEC 25010:2011] [ISO 9241-210:2010] Zhang et al. (2021)	X		X	X
Data Management	Ensures efficient and optimized handling of agricultural data collected via IoT sensors and edge computing.	[ISO/IEC 25010:2011] [ISO/IEC 12207:2017] Simplified	X	X	X	X
Reliability & Scalability	Ensures the system works across varying conditions and can scale as needed.	[IEEE 1012-2016] [ISO/IEC 25010:2011]	X	X	X	X
Efficiency	Ensures the system operates with minimal resource consumption and delivers quick, actionable insights in real-time.	[ISO/IEC 25010:2011] [ISO/IEC 27001:2013]	X	X	X	
Accuracy	Ensures the data collected and processed by the system is accurate, enabling precise decisions on irrigation, pest control, and fertilization.	[ISO/IEC 25010:2011] [ISO 27001:2013] extended	X	X	X	X

Definition of Fairness: SQ model

<https://docs.google.com/spreadsheets/d/1FueEFMz3RcKaB2feQTvwa1M5oiWR7xgUDmSjVXYRjOU/edit?gid=0#gid=0>

Learnability	Ensures that farmers can easily learn and use edge computing and IoT systems with minimal training.	[ISO/IEC 25010:2011]	X		X	X
Compatibility	The system needs to connect well with farmer-employed technologies while maintaining operational continuity.	[ISO/IEC 25010:2011],	X		X	X
Cost	Farmers can access the system at a cost-friendly price because operational expenses and infrastructure investment became minimal.	[ISO/IEC 25010:2011],	X	X	X	X

Definition of Fairness: SQ model (cont.)

The SQ model specifies seven fundamental quality characteristics for edge computing in smart agriculture which consist of Data Accessibility, Data Management, Reliability & Scalability, Efficiency, Accuracy, Learnability and Compatibility, as well as Cost. The technical aspects along with environmental characteristics combine with economic requirements to social attributes to form these quality characteristics.Efficiency optimizes resource utilization and decreases environmental effect and expenses while benefiting technical usage with better resource optimization. Time-efficient data processing through this system leads to fair decisions being made promptly. Learnability enables farmers to easily take up the system so the approach promotes fairness within society.

The mapping of cost criteria to the economic facet makes the technology affordable to farmers thereby guaranteeing both economic fairness and wide technological accessibility.The system achieves excellent efficiency and scalability and operational reliability through standards ISO/IEC 25010 and IEEE 1012 while providing equal technology access and enhancing resource management capabilities alongside sustainable development across all dimensions.

Fairness perspective

	Description
Desired Quality	Edge computing solutions in agriculture provide fair resource distribution to users delivering sustainable farming especially to rural farmers across India. Through this system users obtain access to data alongside improved resource efficiency and live decision capabilities.
Concerns	Data Accessibility, Accuracy and Efficiency
Architectural Tactics	1. Local Data Processing via Edge Computing 2. IoT Sensor Network for Real-Time Data Collection 3. Distributed Edge Devices 4. Blockchain for Transparency
Possible KPIs	1. Data Accuracy Rate 2. Resource Optimization 3. System Uptime and Reliability 4. Farmer Adoption Rate
Problems and Pitfalls	1. Data Inaccuracy from sensors 2. Connectivity Issues 3. Hardware Limitations 4. Farmer Education and Trust 5. Scalability challenges

Overview of edge computing in the agricultural Internet of Things: key technologies, applications, challenges, (2020), IEEE

Definition of KPI*

Real-time Data Processing and Decision-Making Efficiency for smart agriculture					
Goal	**Critical Success Factor (CSF)**	**Key Performance Indicator (KPI)**		**Metric(s)**	**Measure(s)**
Agricultural operations should become more efficient for rural farmers in India through the provision of accessible real-time data systems which incorporate edge computing and IoT sensors.	Data Accessibility, real-time decision-making, and efficiency for smallholder farmers.	Reduction in latency for data processing, improving real-time decision-making that enhances crop yield, resource efficiency, and pest control. Percentage = f(latency reduction, crop yield improvement, resource optimization)		**Latency**: The time taken for real-time data (e.g., irrigation or pest control updates) to be processed at the edge compared to cloud processing. **Data Accessibility**: Evaluate the percentage of the farm area covered by edge devices and how quickly data is accessible to farmers. **Efficiency**: Monitor the efficiency of irrigation, pest control, and fertilizer application after system integration.	**Latency Measurement**: Time (in seconds) taken for data from edge devices to be processed and used for decision-making. **Coverage Area**: Percentage of the farm area with IoT sensor coverage. **Yield Improvement**: Measure improvements in crop yield and reductions in resource waste (e.g., water, pesticides, and fertilizers). **System Downtime**: The downtime of edge computing nodes, ensuring continuous system operation with minimal interruptions. **Processing Time**: The time taken for data processing at the edge for irrigation, pest control, and other critical decisions.
		Target	**Action(s)**		
		Increase real-time data processing at the edge by 40%, reduce cloud dependency by 30%, and improve overall crop yield by 15% within the next farming cycle.	Technological infrastructure gets installed in rural territories to operate edge computing equipment along with IoT sensors for immediate data operations. Farmer Training: The system should provide education to farmers about using time-sensitive data to enhance their irrigation management and pest management and fertilizer use strategies. A continuous performance assessment and edge computing optimization process should run to minimize delay times.		

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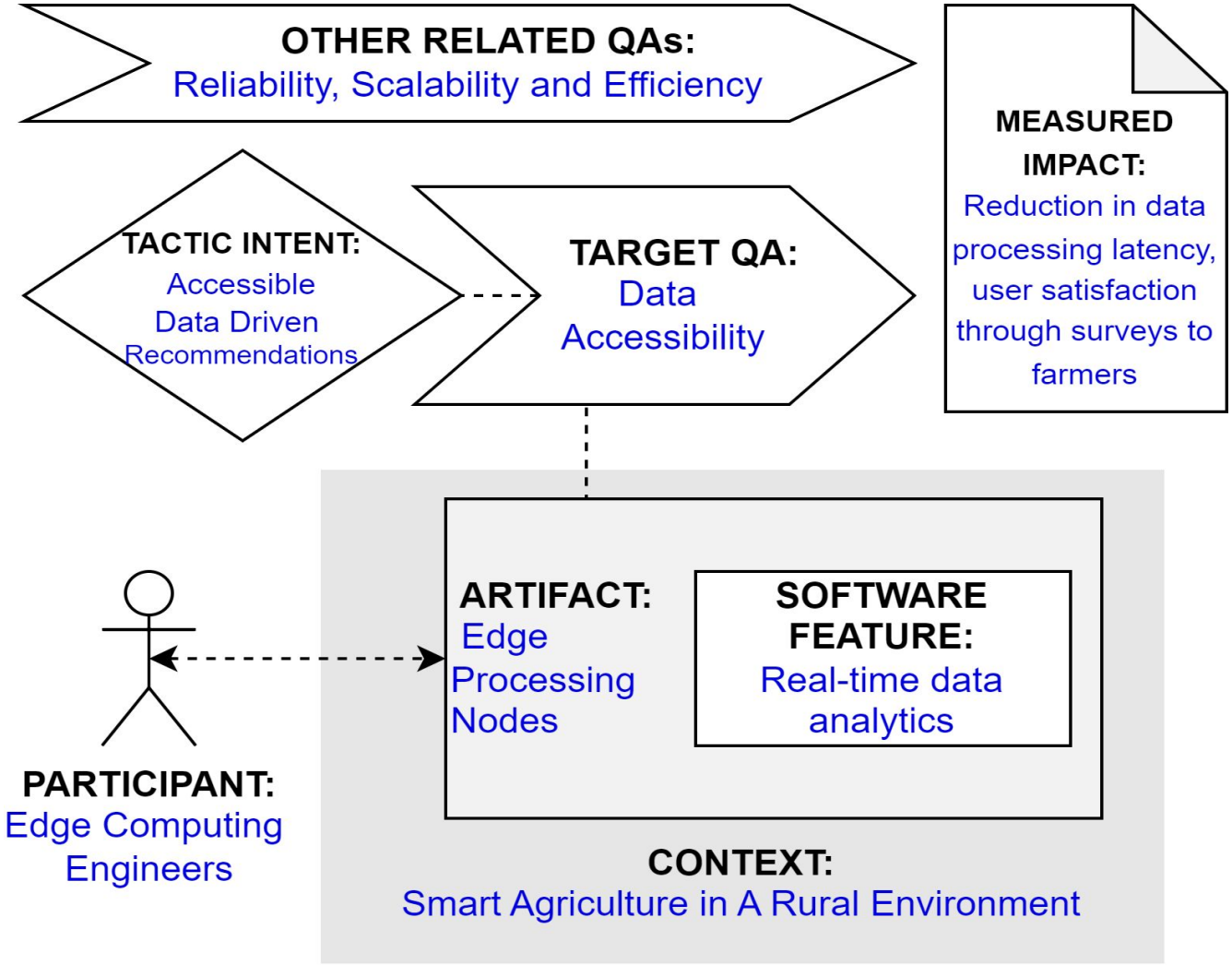
ISO/IEC 25010:2011 and ISO/IEC 12207:2017 extended and simplified

CON-PI: a distributed Container-Based Edge and Fog computing framework. (2022, March 15). IEEE Journals & Magazine | IEEE Xplore. https://ieeexplore.ieee.org/document/9508413

Architectural Tactic(s) for Fairness*

title: "Local Data Processing For Equitable Decision Making"
tags: “edge-computing decision-making local-processing”
t-sort: "Awesome Tactic"
t-type: "Architectural Tactic"
categories: “Agricultural Data Processing and Management, Edge Computing”
t-description: "Local data processing enables the farmers to make decisions in real time through the analysis of data collected from sensors and devices within the field. By doing this, the system reduces latency and minimizes dependency to centralized cloud resources which means that small scale farmers will have access to actionable insights on irrigation, fertilization and pest control reducing costs within a timely manner. Additionally, the intuitive dashboard displays all the necessary insights with the relevant information.”
t-participant: "Edge Computing Engineers"
t-artifact: "Edge Processing Nodes"
t-context: "Smart Agriculture in A Rural Environment."
t-feature: "Real Time Data Analytics"
t-intent: "To ensure that all farmers, regardless of connectivity or infrastructure limitations, have equal access to reliable data-driven recommendations "
t-targetQA: "Data Accessibility"
t-relatedQA: "Reliability & Scalability, Efficiency"
t-measuredimpact: "Reduction in data processing latency (30% decrease compared to centralized cloud), increased user satisfaction through surveys to farmers"
t-source: "Sathya, D., Thangamani, R., & Balaji, B. S. (2024). The Revolution of Edge Computing in Smart Farming. In Intelligent Robots and Drones for Precision Agriculture (pp. 351-389). Cham: Springer Nature Switzerland."
t-source-doi: "https://doi.org/10.1007/978-3-031-51195-0_17"

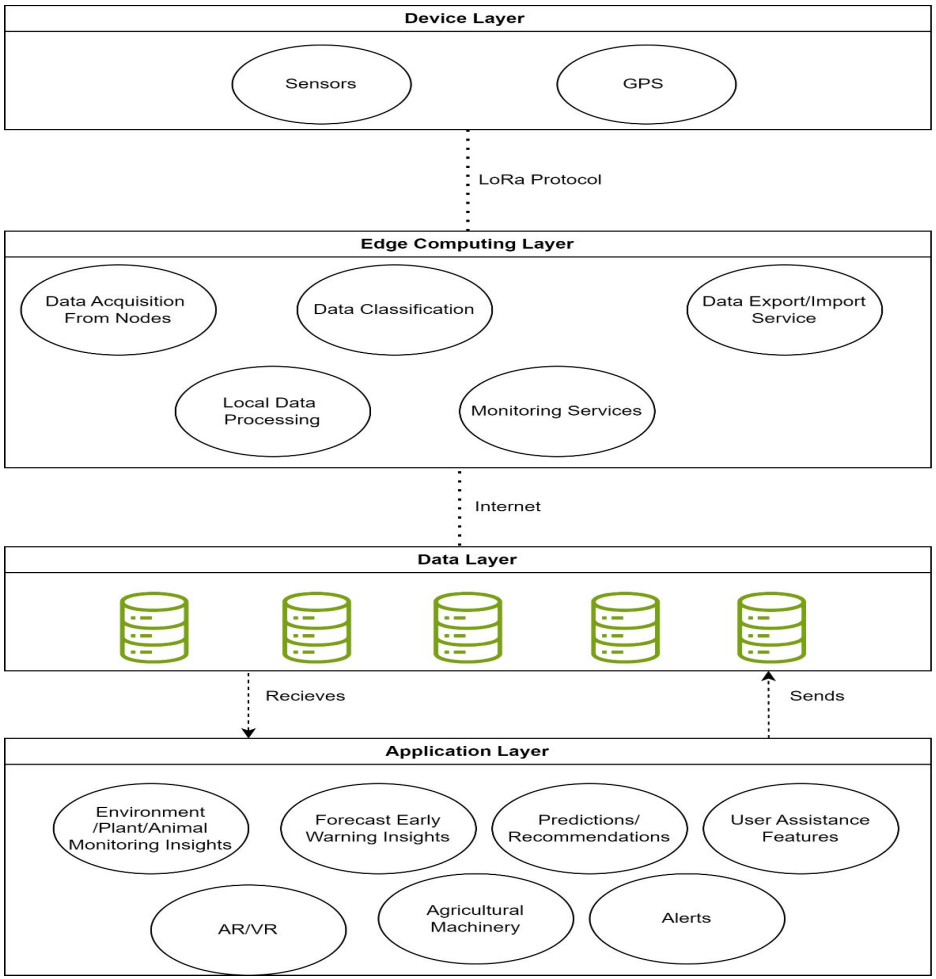
Architectural Tactic(s) for Fairness* Continued

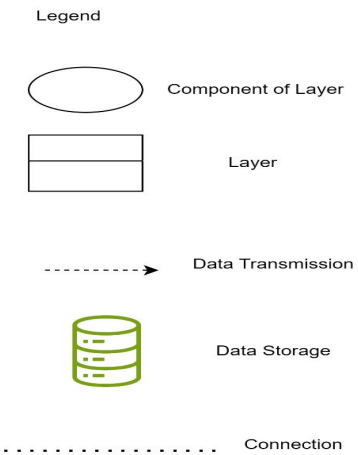


Template: P. Lago, R. v.d. Wiel, Awesome and Dark Tactics, online catalog at http://s2group.cs.vu.nl/AwesomeAndDarkTactics

Fairness view: <Local Data Processing View>

https://drive.google.com/file/d/1SPKJirnQk9UvQGN8H7xKfwCSQASrUYj-/view?usp=sharing





Rathore, N., Naik, R. R., Gautam, S., Nahta, R., & Jain, S. (2024, January). Smart Farming Based on IoT-Edge Computing: Exploiting Microservices' Architecture for Service Decomposition. In International Conference on Data Management, Analytics & Innovation (pp. 425-437). Singapore: Springer Nature Singapore.
Zhang, Xihai & Cao, Zhanyuan & Dong, Wenbin. (2020). Overview of Edge Computing in the Agricultural Internet of Things: Key Technologies, Applications, Challenges. IEEE Access. 8. 1-1. 10.1109/ACCESS.2020.3013005.

Fairness view (cont.)

The device layer makes use of various sensor and GPS nodes deployed all throughout the farm. The data (soil moisture, temperature, crop health, humidity, machinery, location, weather) acquired by these sensors and the GPS nodes are connected to the edge nodes in the edge computing layer through wireless transmission via the LoRa module. Edge nodes are considered to be the devices that act as intermediaries between the device layer and the edge service layer. Since they are positioned closer to the edge devices, edge nodes aggregate and preprocess data locally. They promote efficient decision-making and provide a buffer for managing data before sending it to the data layer.These nodes perform data acquisition from device layer nodes, local data processing (filtering and summarizing data locally), data classification which groups data from the different sensors, monitoring service which sensor node status, connection health and prepares to send alerts and data export/import service from edge servers. The processed data is then transmitted to the data layer through Wi-Fi for long term storage. The data layer functions as a vast data pool of all the gathered sensor data and once the user requests for insights regarding any of the components in the application layer, they are able to retrieve them.

Ojuawo, Olutayo & Jiboku, Folahan. (2023). Overview of Edge Computing and Its Significance in the Era of IoT and Big Data.